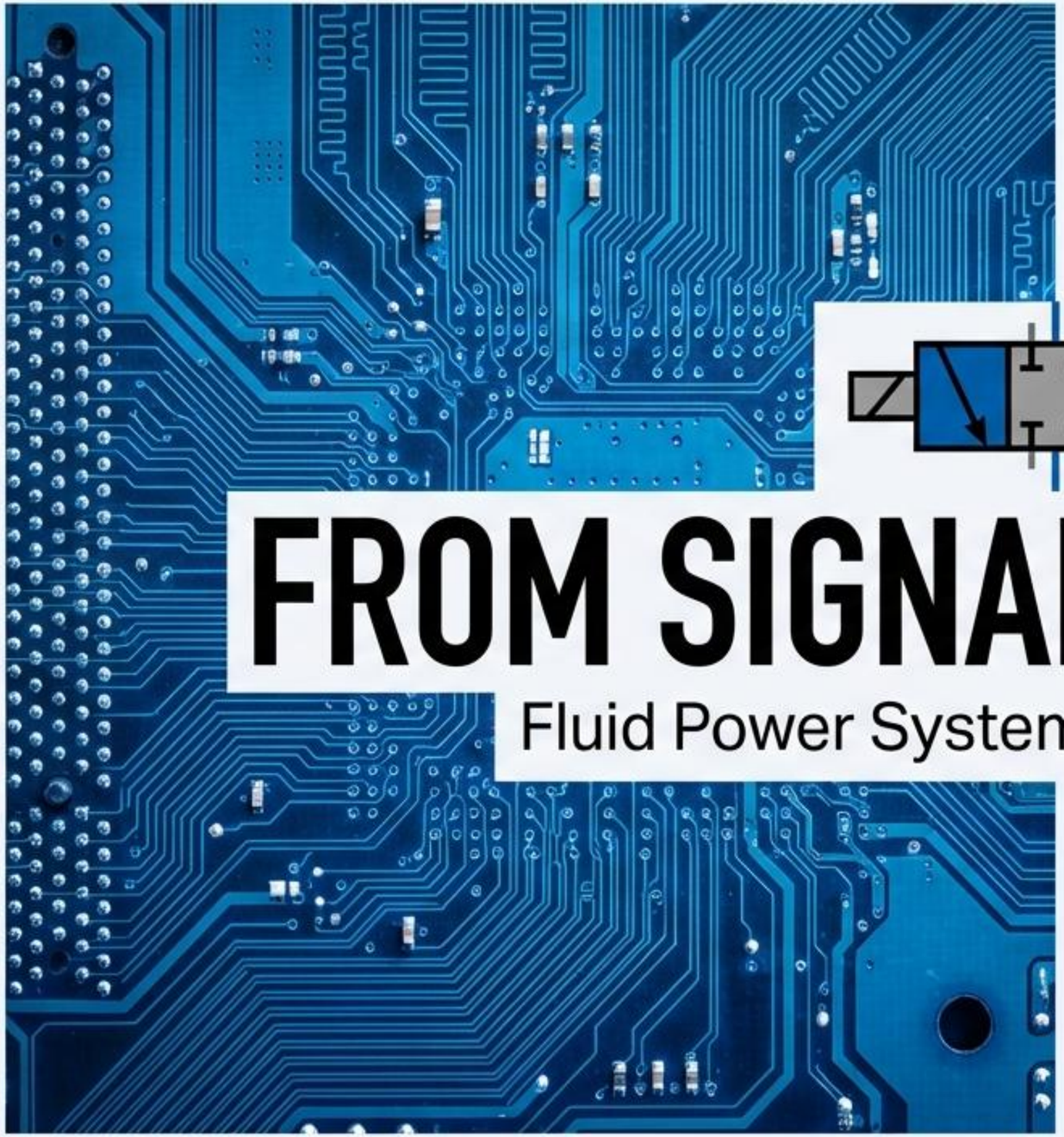


FROM SIGNAL TO MOTION

Fluid Power Systems & Mechatronics

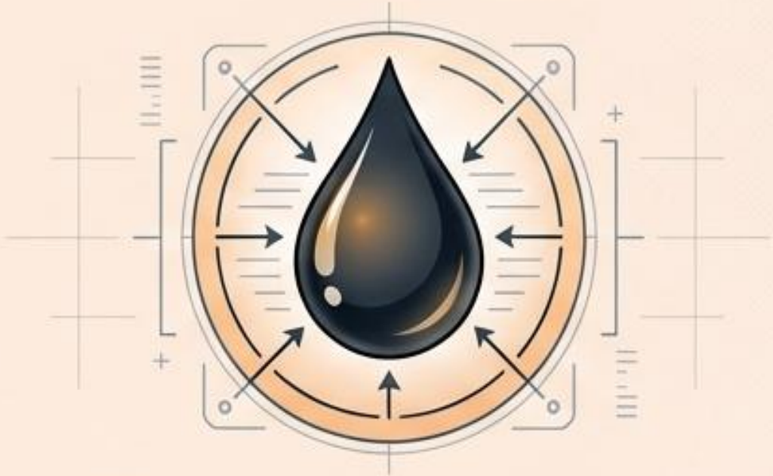
Based on Chapter 7: Mechatronics by William Bolton

PNEUMATICS

- Compressible (Spongy)
- Low Viscosity
- Exhausts to Atmosphere
- Low Force / High Speed

THE FAST & COMPLIANT

HYDRAULICS

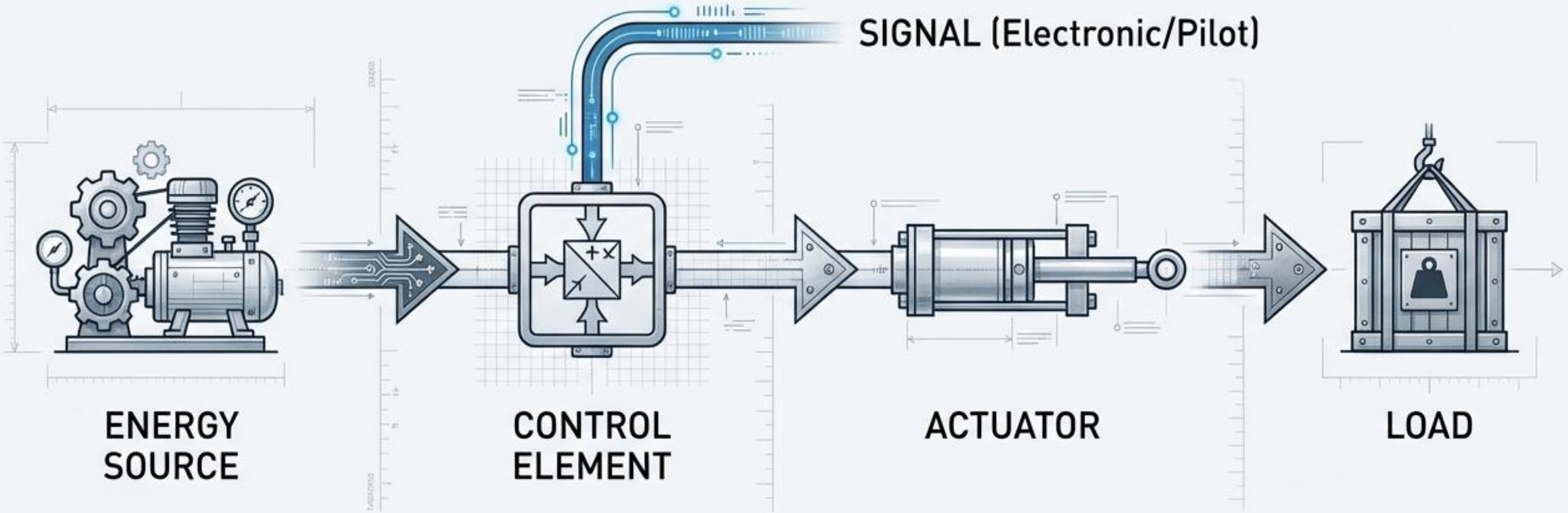


- Incompressible (Stiff)
- High Viscosity
- Return Lines Required
- High Force / Precision

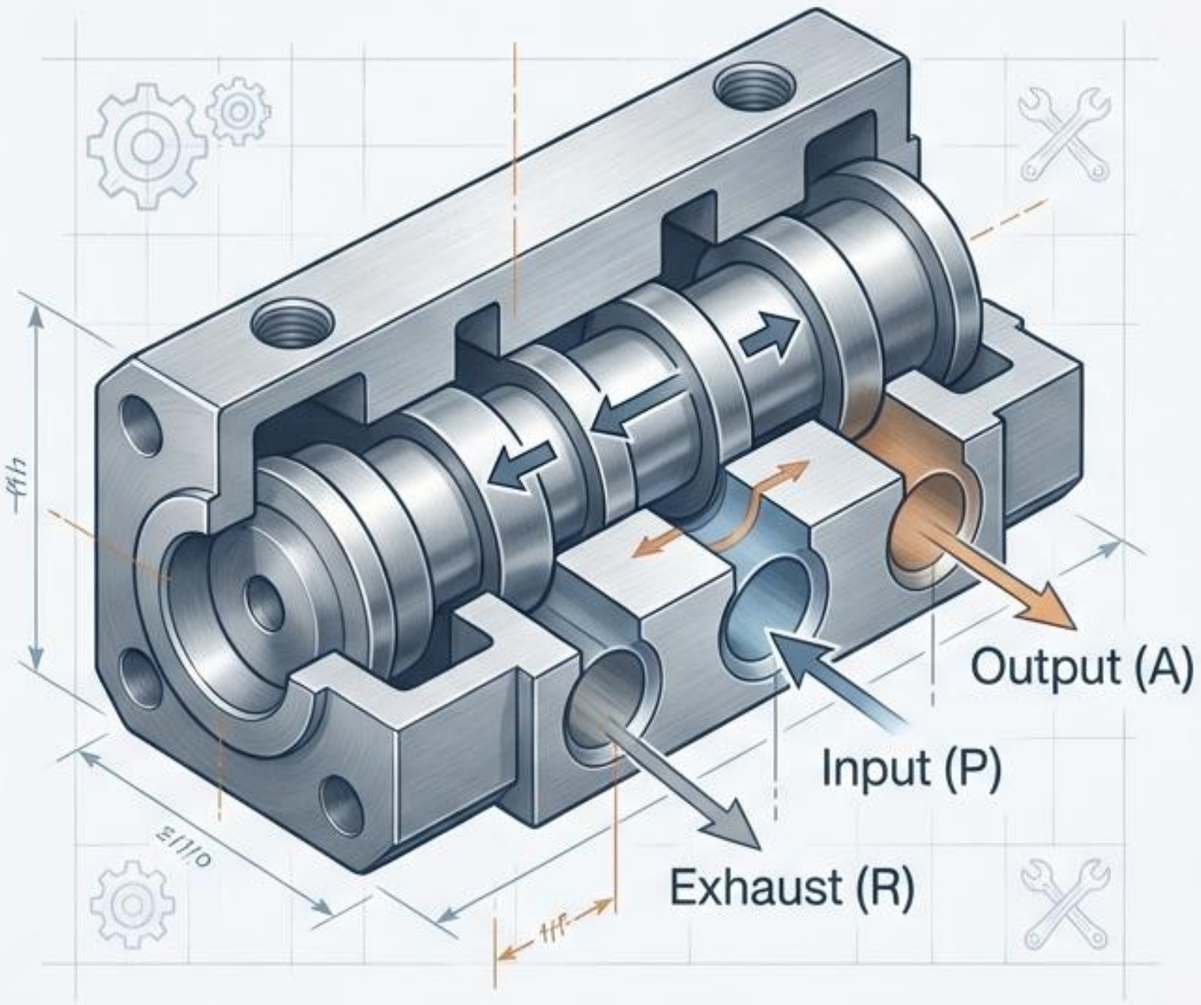
THE STRONG & PRECISE

Different Mediums, Identical Control Logic

ACTUATION ARCHITECTURE



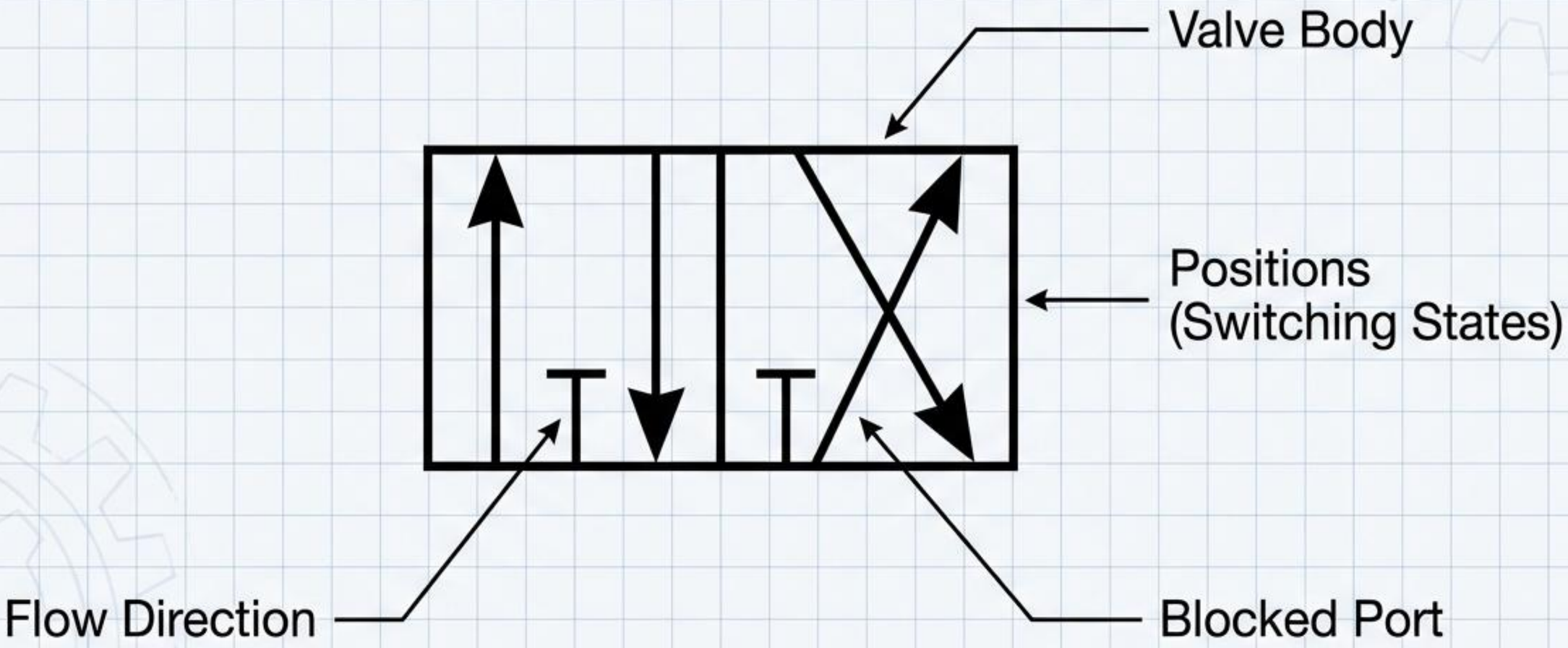
DIRECTIONAL CONTROL VALVES (THE ROUTER)



- **Function:** Directs fluid to extend/retract actuators.
- **Mechanism:** Sliding spool covers/uncovers ports.
- **Actuation:** Driven by Solenoid, Spring, or Pilot Air.

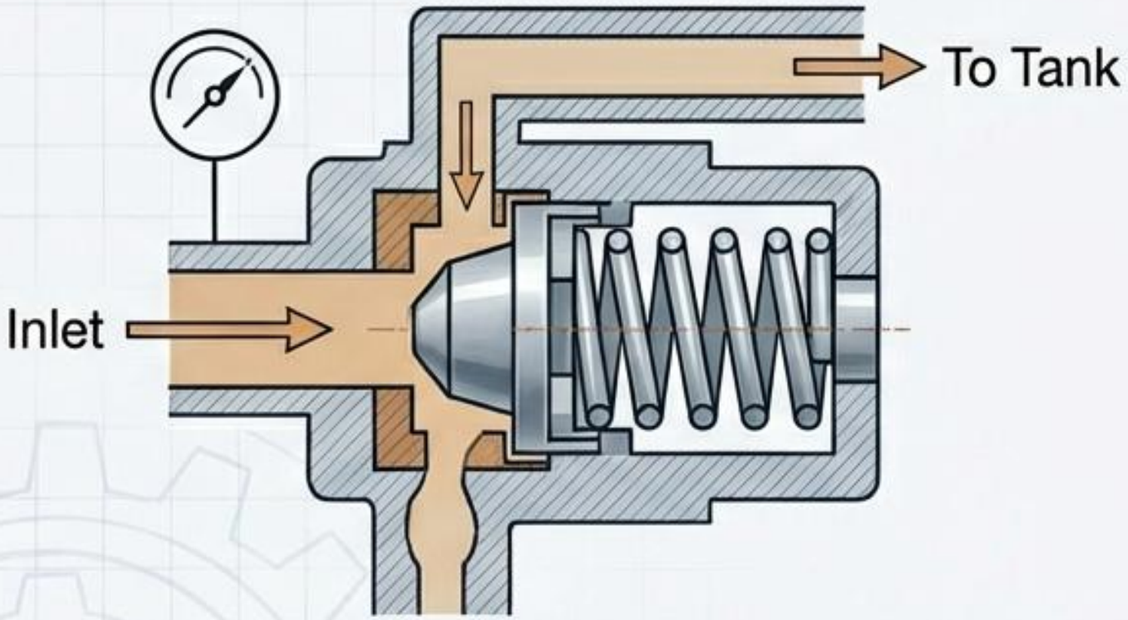
SPEAKING THE LANGUAGE: ISO SYMBOLS

Decoding the 4/2 Directional Valve



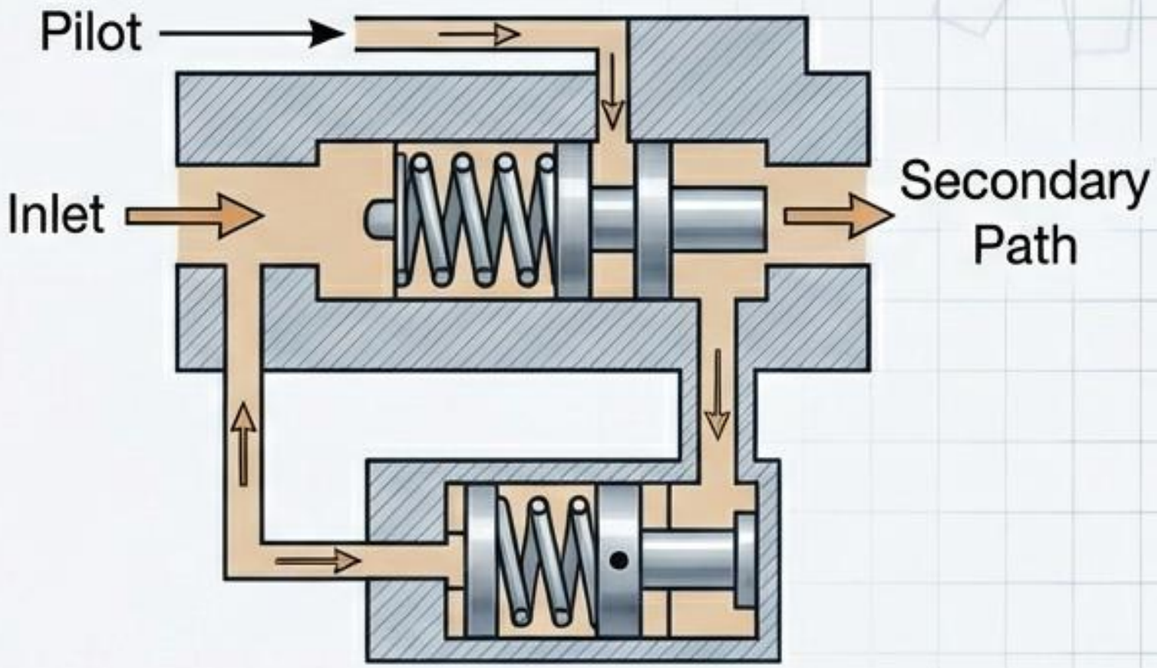
PRESSURE CONTROL VALVES

PRESSURE RELIEF VALVE



THE PROTECTOR (Safety limit)

SEQUENCE VALVE

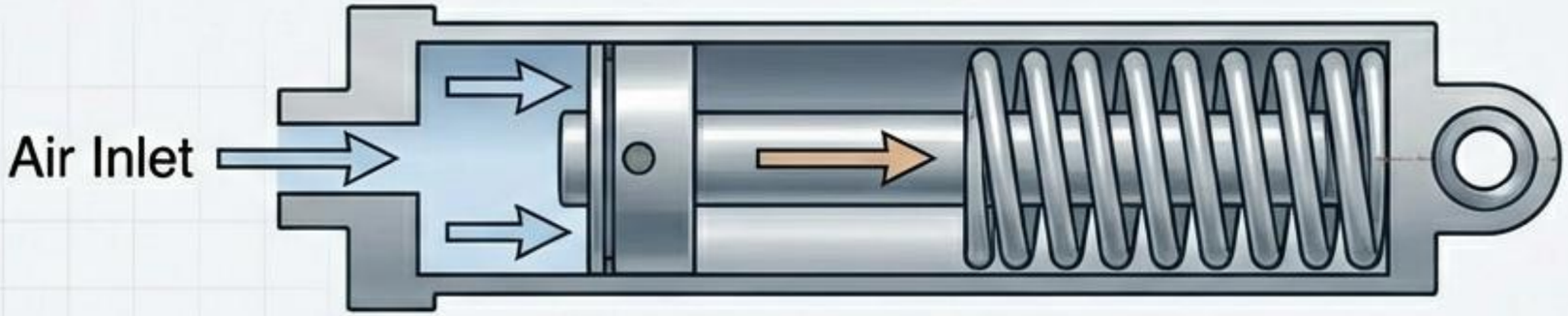


THE SEQUENCER (Logic)

Use Case: Clamp the workpiece first (pressure rise), then drill (sequence opens).

LINEAR ACTUATORS: THE MUSCLE

Single-Acting Cylinder



Power Extend, Spring Return

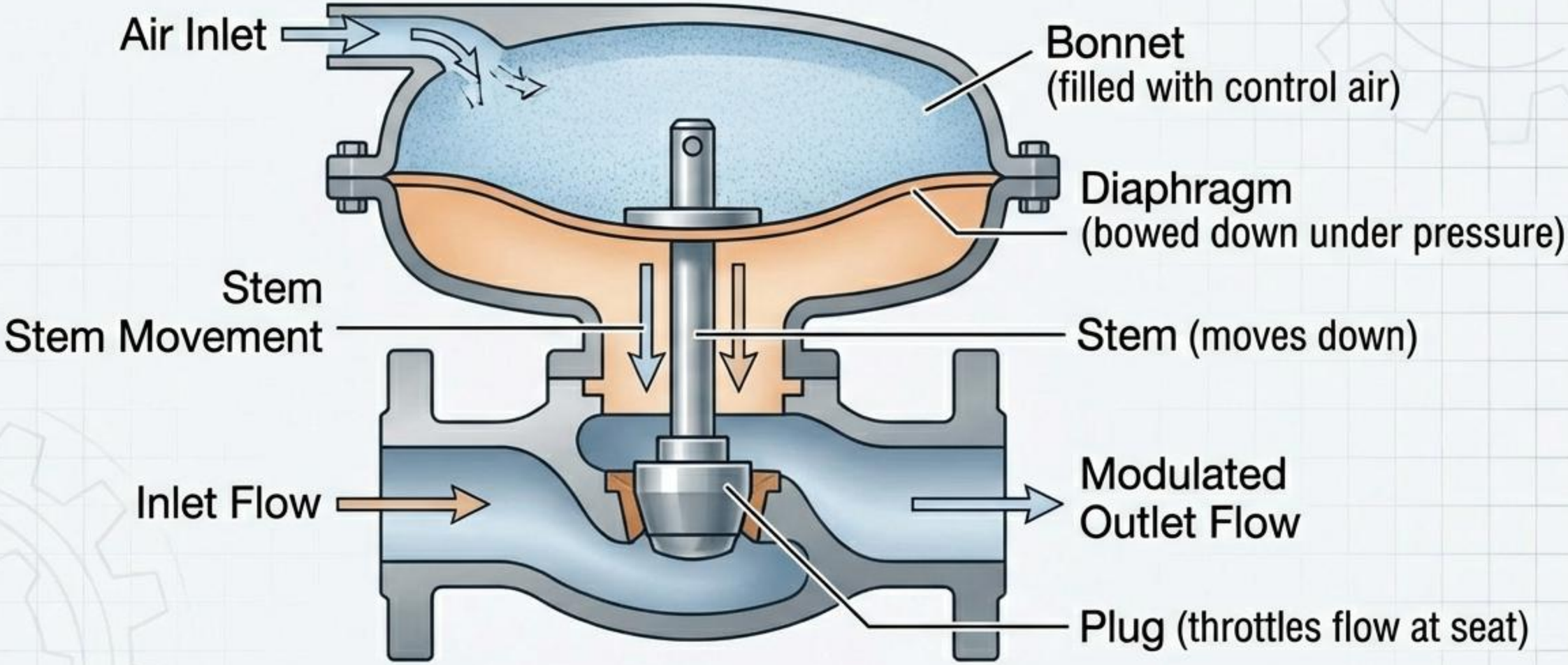
Double-Acting Cylinder

Power Extend, Power Retract

Force = Pressure × Area

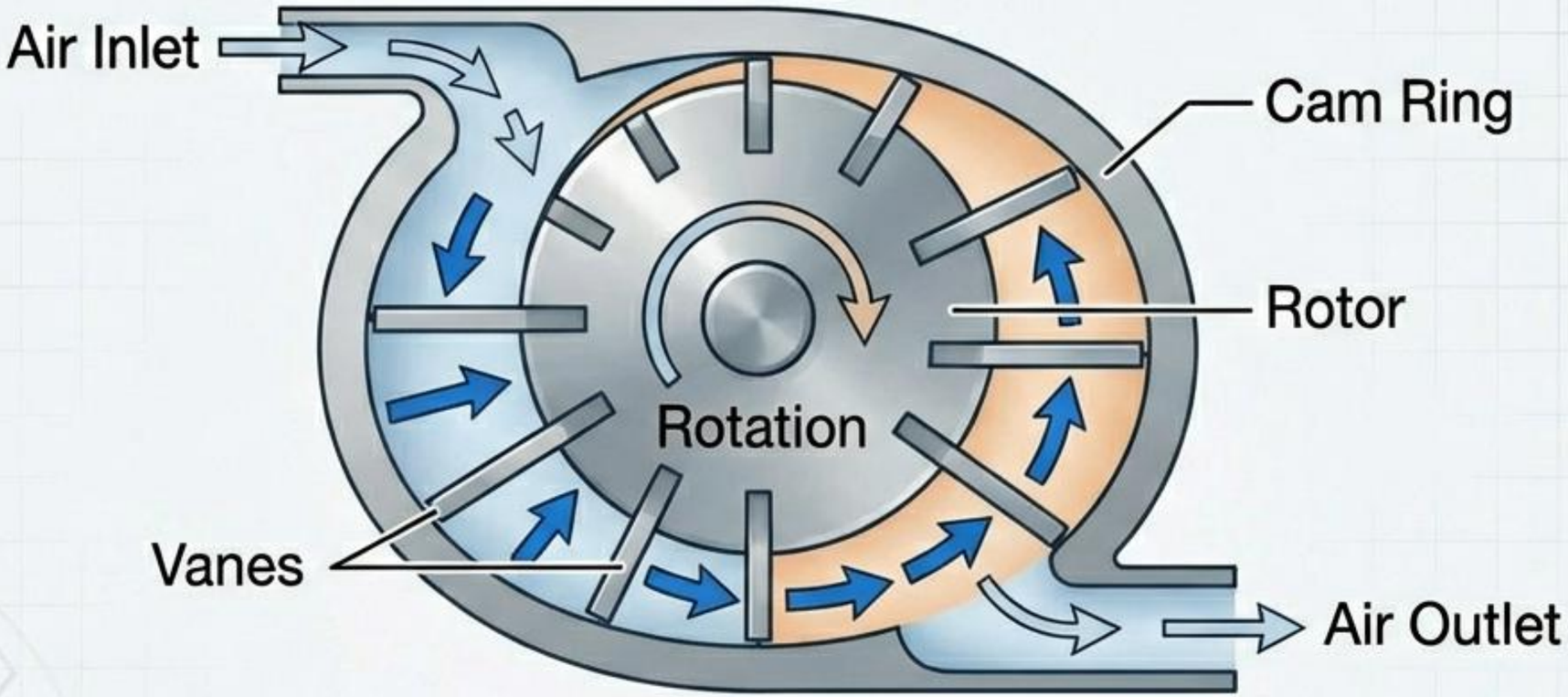
Note: Retraction force is lower due to rod area.

PROCESS CONTROL: FLOW MODULATION



Used to throttle flow rate (variable control), not just On/Off switching.

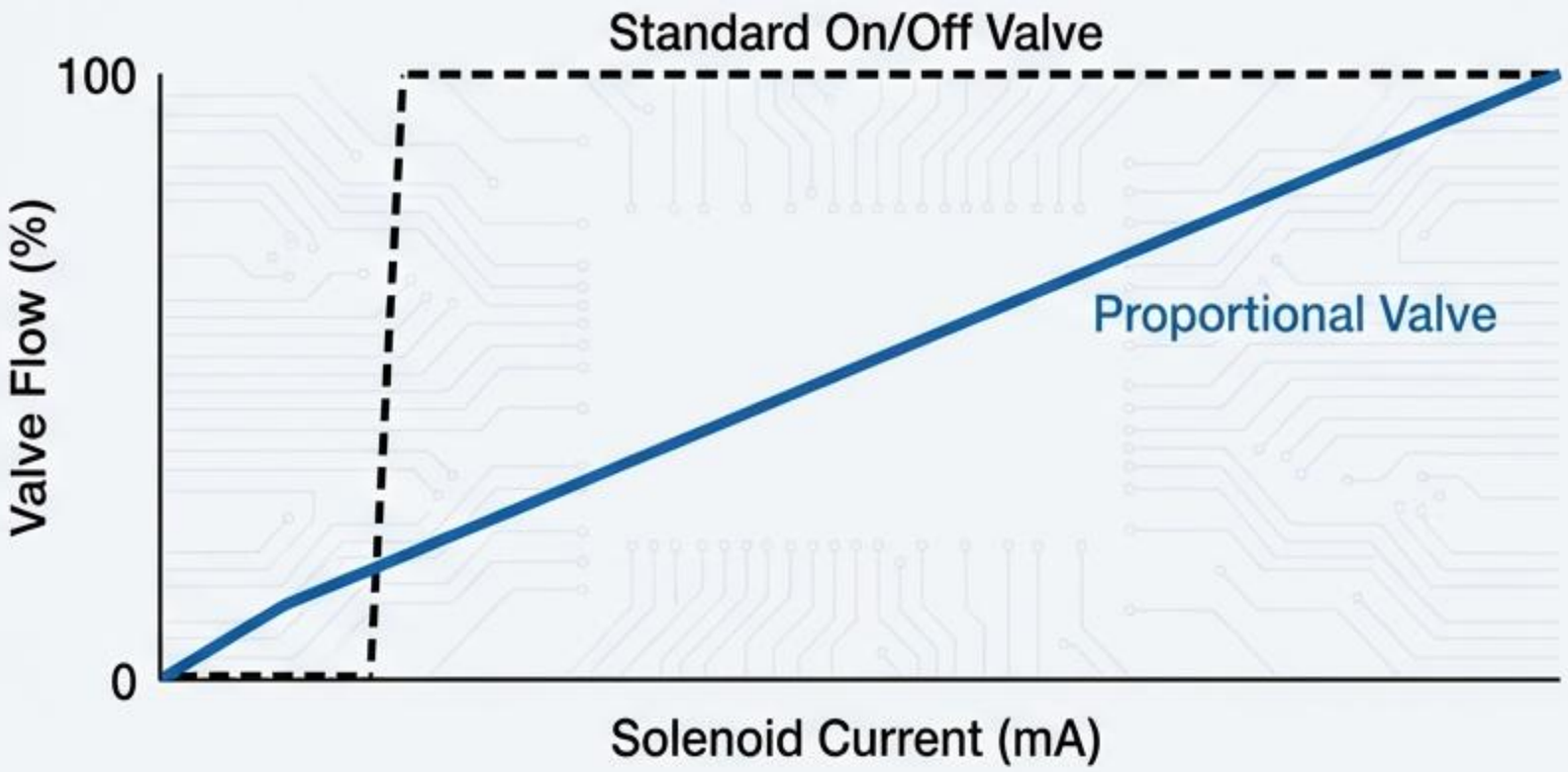
ROTARY ACTUATORS



Converting fluid pressure directly into Torque and RPM.

The pneumatic equivalent of an electric motor.

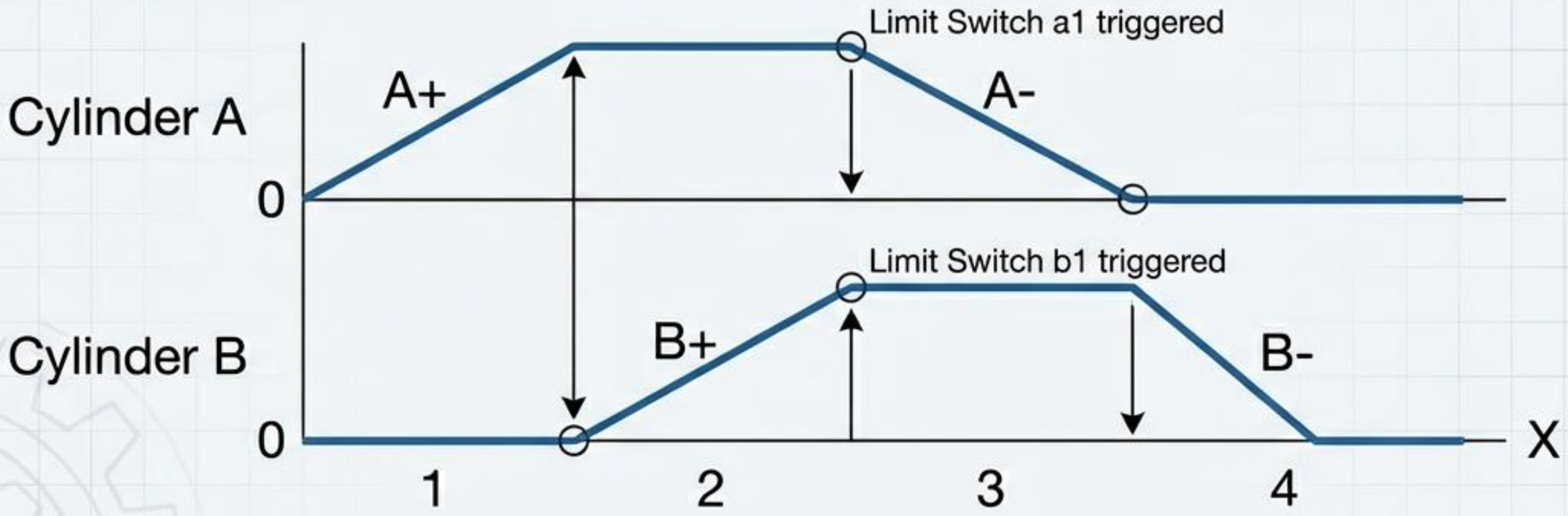
SERVO & PROPORTIONAL VALVES



Bridging the gap between electronic precision and fluid power. Proportional valves allow variable speed and force control based on current magnitude.

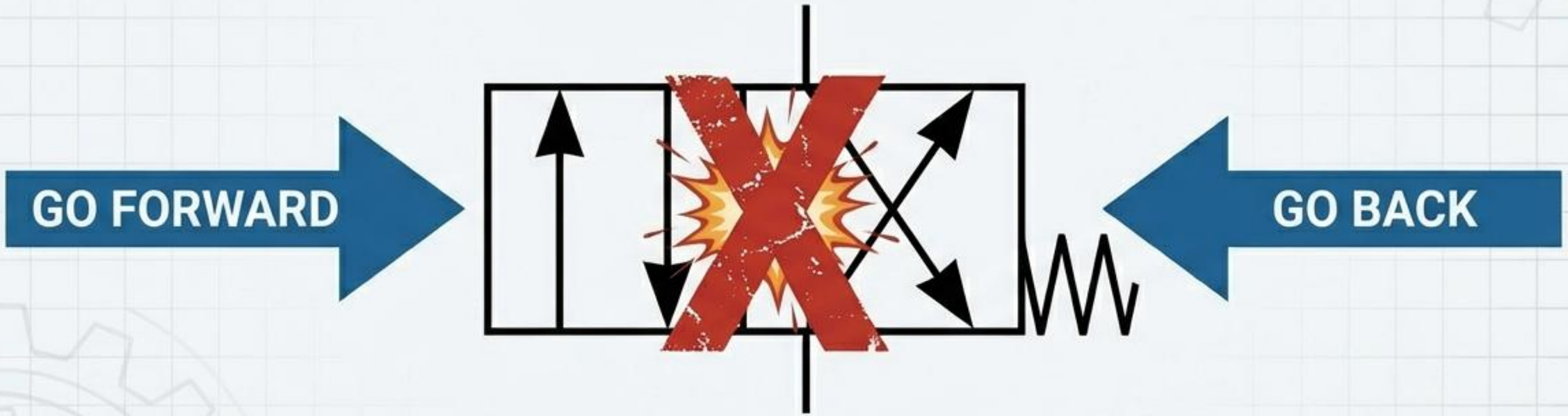
SEQUENTIAL CONTROL SYSTEMS

Displacement-Step Diagram



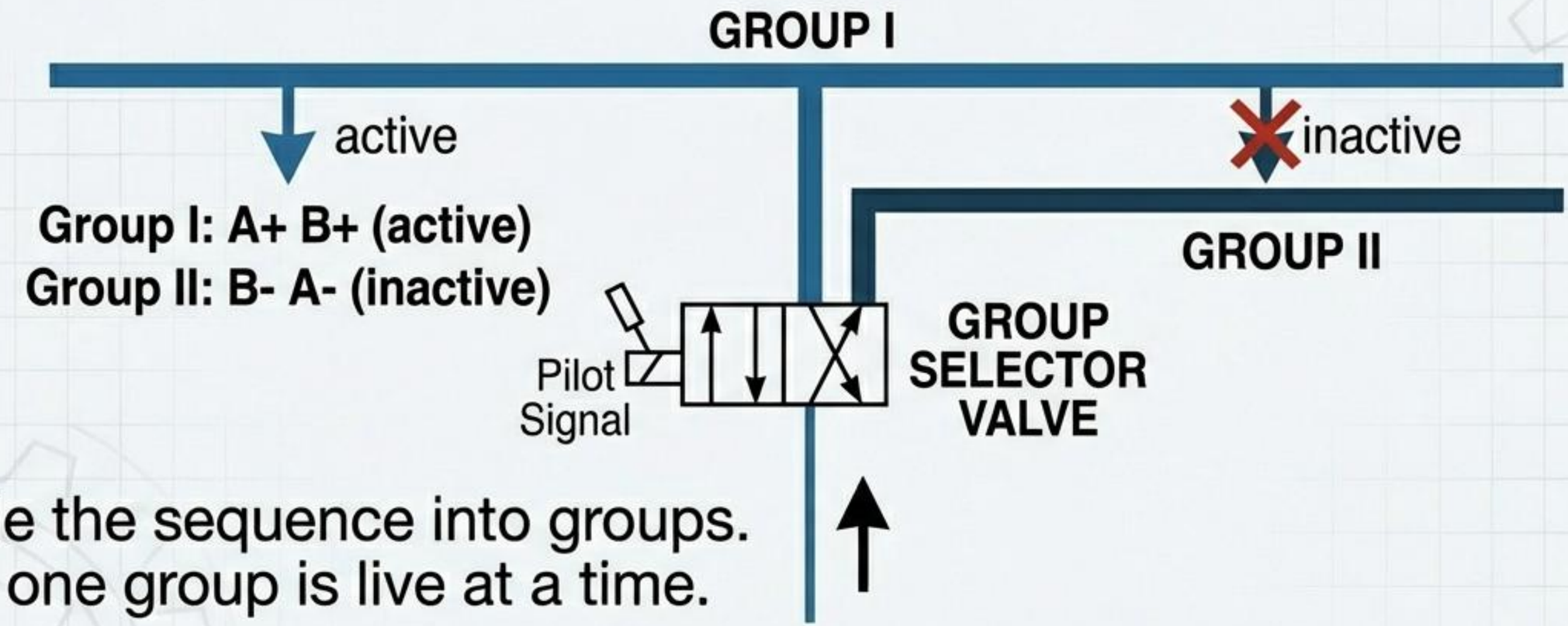
Automating a series of movements using physical feedback (limit switches) to trigger the next step.

THE PROBLEM: SIGNAL CONFLICT



In complex sequences (like A+ B+ B- A-), a valve may receive extend and retract signals at the same time. This “Trapped Signal” causes the system to stall.

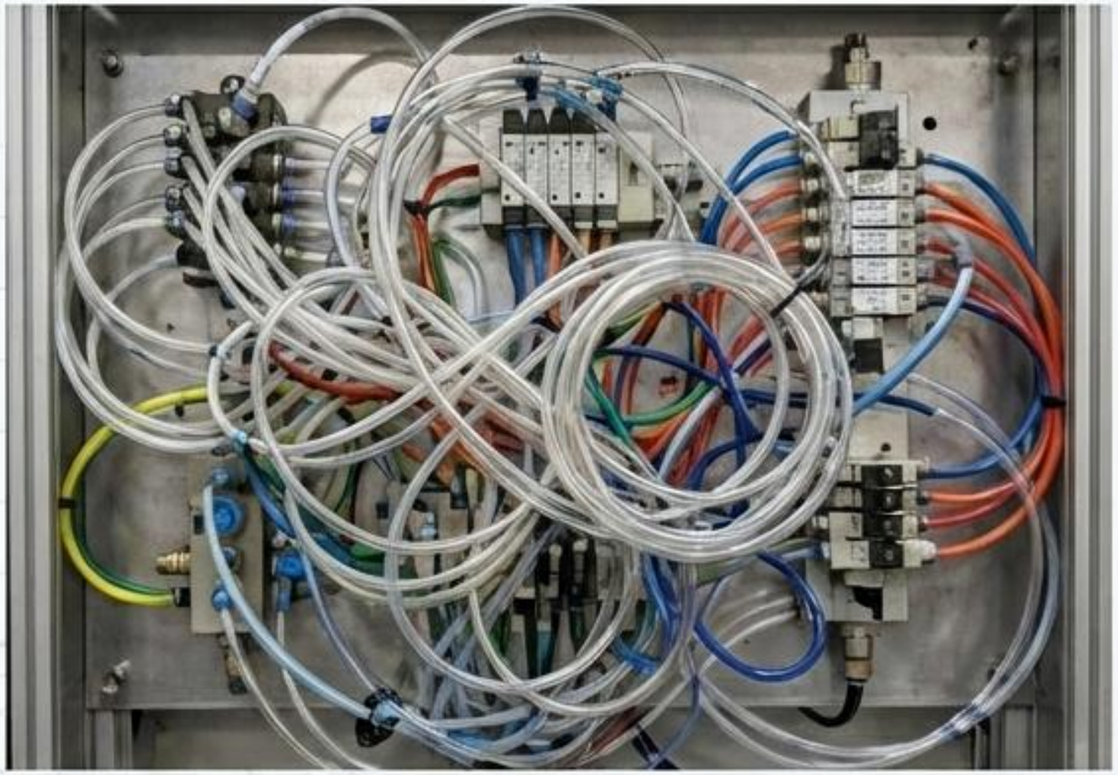
THE SOLUTION: CASCADE CONTROL



Divide the sequence into groups.
Only one group is live at a time.

This visualization demonstrates how physically disconnecting the supply prevents signal conflict.

EVOLUTION: HARD-WIRED VS. PLC CONTROL



Pure Fluid Logic (Hard-Wired)

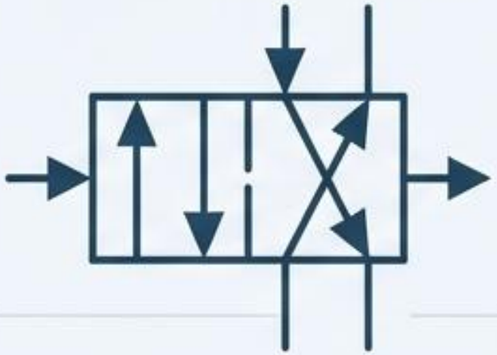
Complex, inflexible, difficult to troubleshoot.

PLC Control (Software Logic)

Flexible, programmable, simplified plumbing.

KEY TAKEAWAYS

Fluid Power: High force (Hydraulic) or High speed (Pneumatic).



Valves: The logic gates controlling Direction, Pressure, and Flow.

Actuators: Converting fluid energy into mechanical motion.

Control: From mechanical sequencing to electronic PLCs.

"Mechatronics is the art of integrating the brain (electronics) with the muscle (fluid power)."